

Design and Analysis of Algorithm

Lecture 26: Traversal

Content

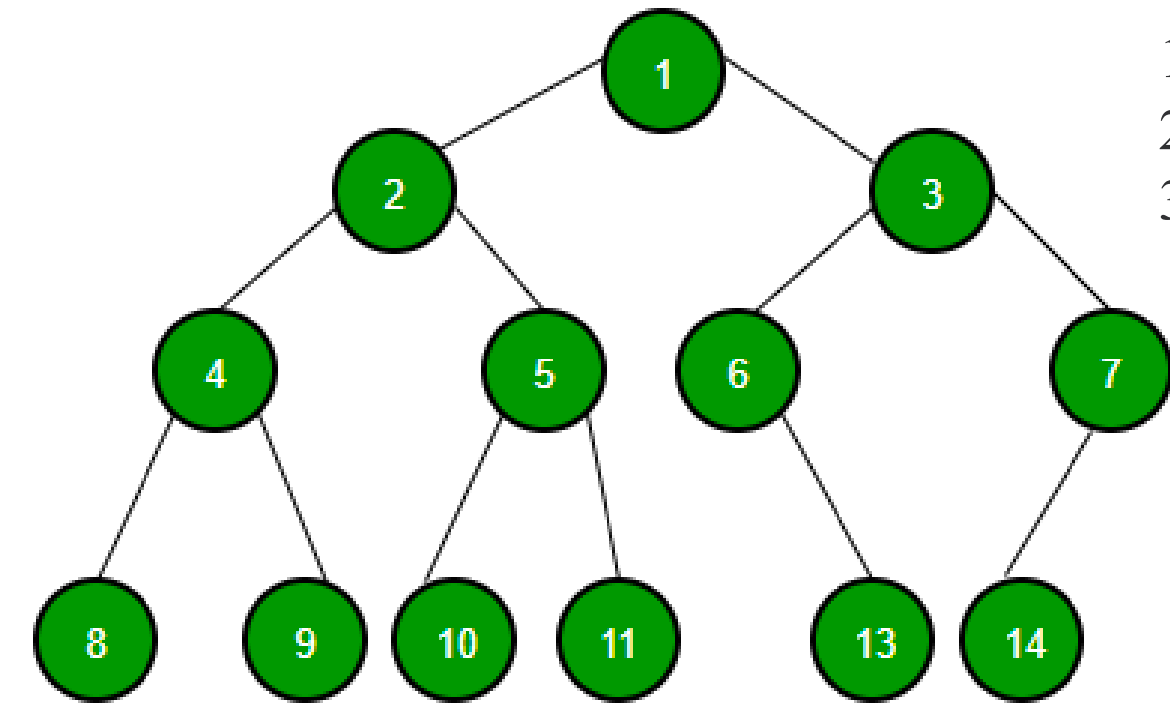


- **Tree Traversal**

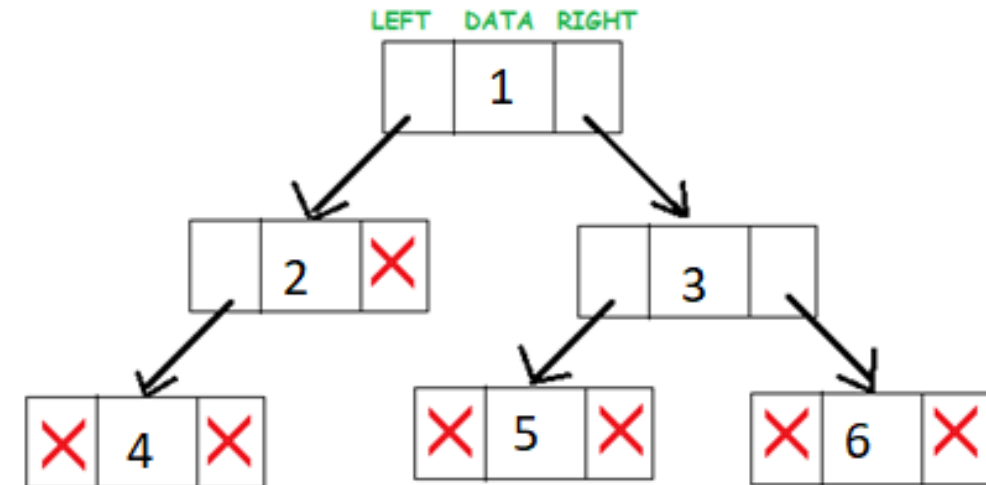
Binary Tree

Definition:

A binary tree is a hierarchical data structure in which each node has at most two children generally referred as left child and right child.

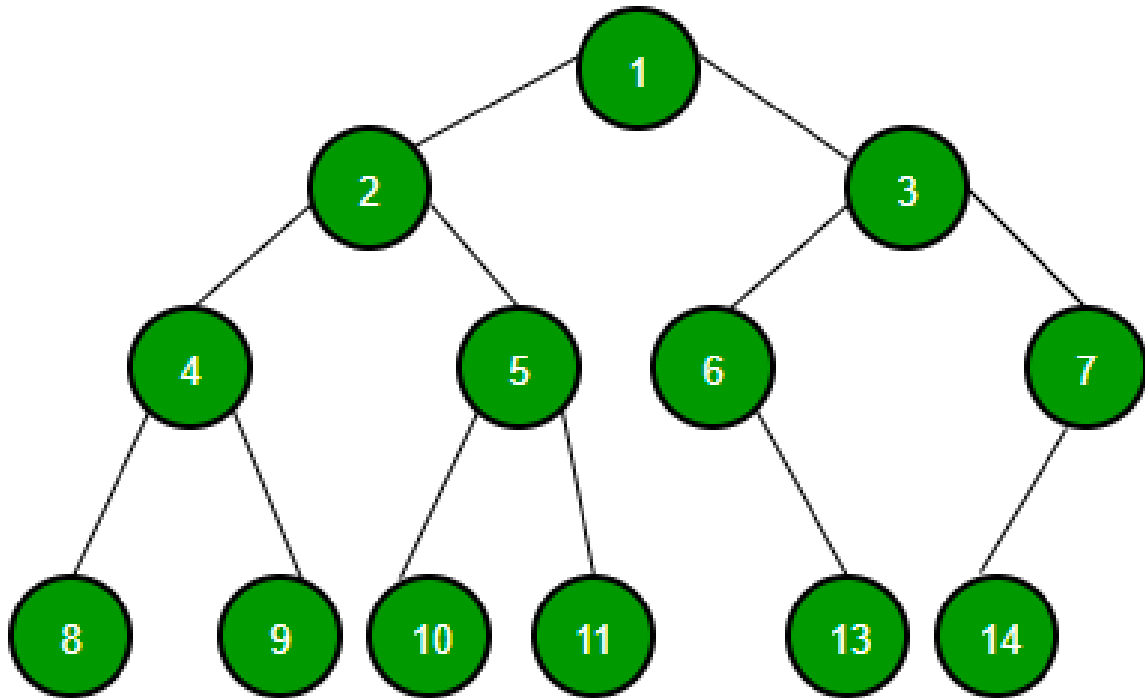


1. Data element
2. Pointer to left subtree
3. Pointer to right subtree



Binary Tree

Binary trees are mainly used for searching and sorting as they provide a means to store data hierarchically. Some common operations that can be conducted on binary trees include insertion, deletion, and traversal.



Node.left $\Rightarrow$ left node

Node.right $\Rightarrow$ right node

Node.parent $\Rightarrow$ parent node

Tree: Traversal

When the search necessarily involves the examination of every vertex in the object being searched, it is called a traversal.

Traversal Techniques

- Preorder
- Inorder
- Postorder

Tree Traversal: **Inorder**

Start at the root

Visit the left child of each node, then the node, then any remaining nodes

Recursive algorithm for inorder traversal:

- If tree is not empty,

 - Perform inorder traversal of left subtree of root

 - Visit root node of tree

 - Perform inorder traversal of its right subtree

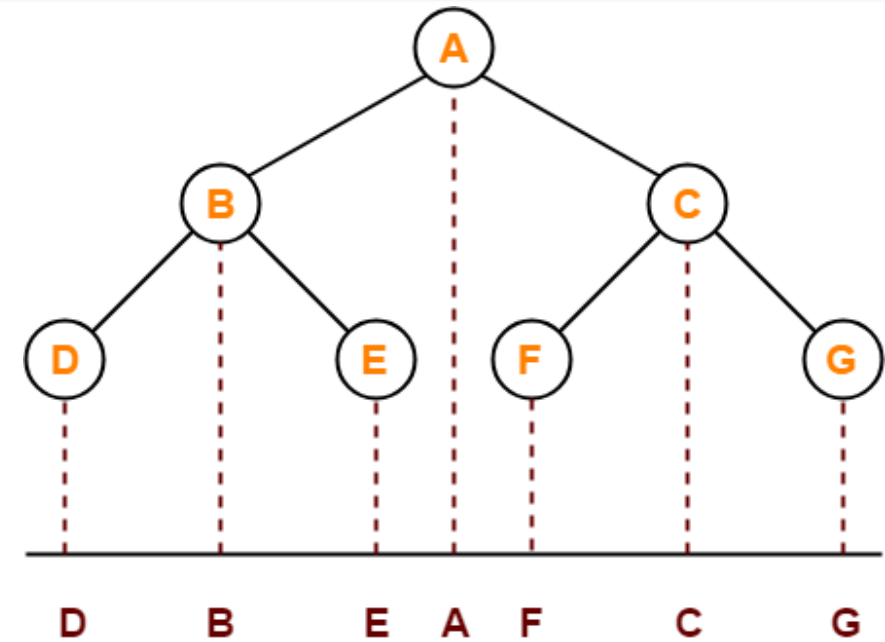
Tree Traversal: Inorder

Approach

In this traversal method, the left subtree is visited first, then the root and later the right sub-tree.

```

1  Algorithm InOrder(t)
2  // t is a binary tree. Each node of t has
3  // three fields: lchild, data, and rchild.
4  {
5      if t ≠ 0 then
6      {
7          InOrder(t → lchild);
8          Visit(t);
9          InOrder(t → rchild);
10     }
11 }
```

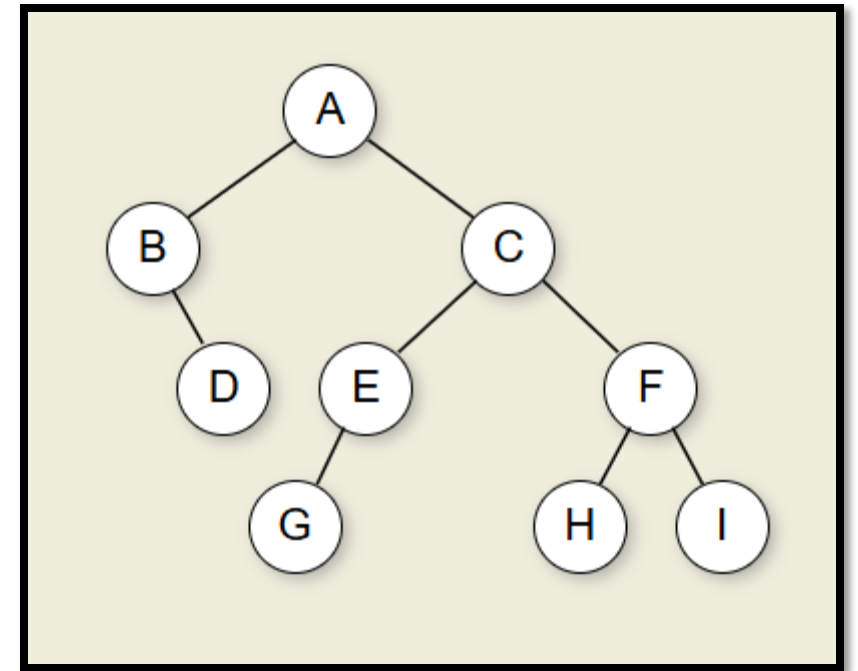


Inorder Traversal : D , B , E , A , F , C , G

Tree Traversal: **Example**

What is the inorder traversal of the following binary tree

The preorder enumeration for the tree is
B D A G E C H F I



Tree Traversal: **Preorder**

Start at the root
Visit each node, followed by its children;
We will choose to visit left child before right

Recursive algorithm for preorder traversal:

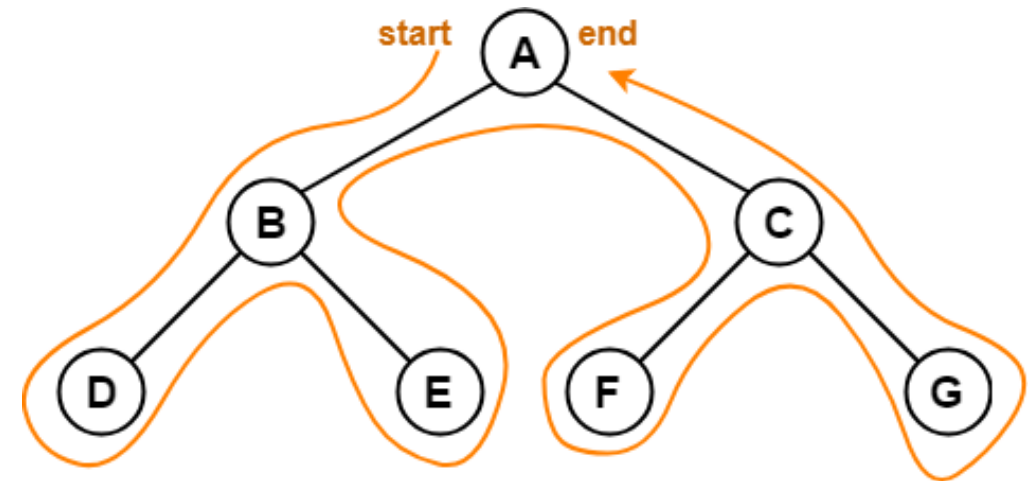
- If tree is not empty,
 - Visit root node of tree
 - Perform preorder traversal of its left subtree
 - Perform preorder traversal of its right subtree

Tree Traversal: Preorder

Approach

In PreOrder traversal, each node is processed before either of its sub-trees. In simpler words, Visit each node before its children.

```
Algorithm PreOrder(t)
// t is a binary tree. Each node of t has
// three fields: lchild, data, and rchild.
{
    if t ≠ 0 then
    {
        Visit(t);
        PreOrder(t → lchild);
        PreOrder(t → rchild);
    }
}
```

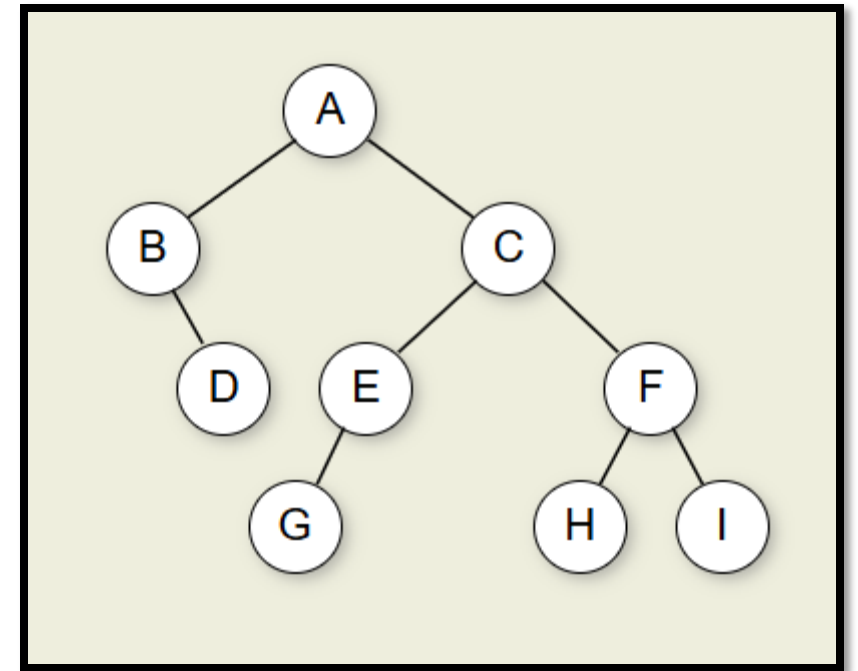


Preorder Traversal : A , B , D , E , C , F , G

Tree Traversal: **Example**

What is the preorder traversal of the following binary tree

The preorder enumeration for the tree is
A B D C E G F H I



Tree Traversal: **Postorder**

Start at the root

Visit the children of each node in the sequence of left to right, then the node

Recursive algorithm for preorder traversal:

- If tree is not empty,

 - Perform postorder traversal of left subtree of root

 - Perform postorder traversal of right subtree of root

 - Visit root node of tree

Tree Traversal: **Postorder**

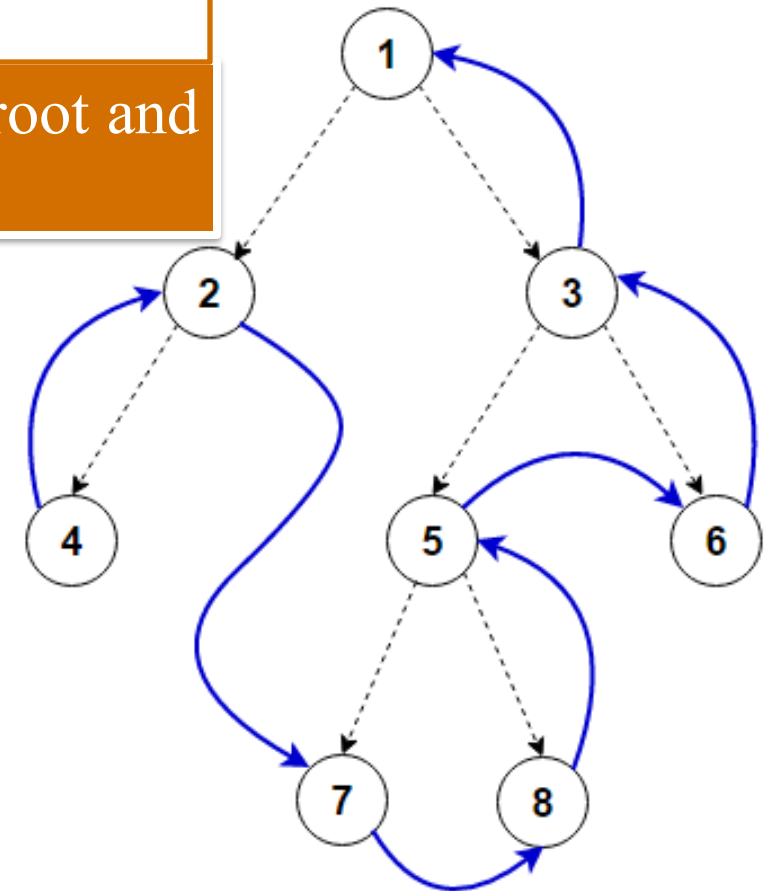
Approach

In this traversal method, the left subtree is visited first, then the root and later the right sub-tree.

Algorithm PostOrder(t)

// t is a binary tree. Each node of t has
// three fields: *lchild*, *data*, and *rchild*.

```
{
  if  $t \neq 0$  then
  {
    PostOrder( $t \rightarrow lchild$ );
    PostOrder( $t \rightarrow rchild$ );
    Visit( $t$ );
  }
}
```

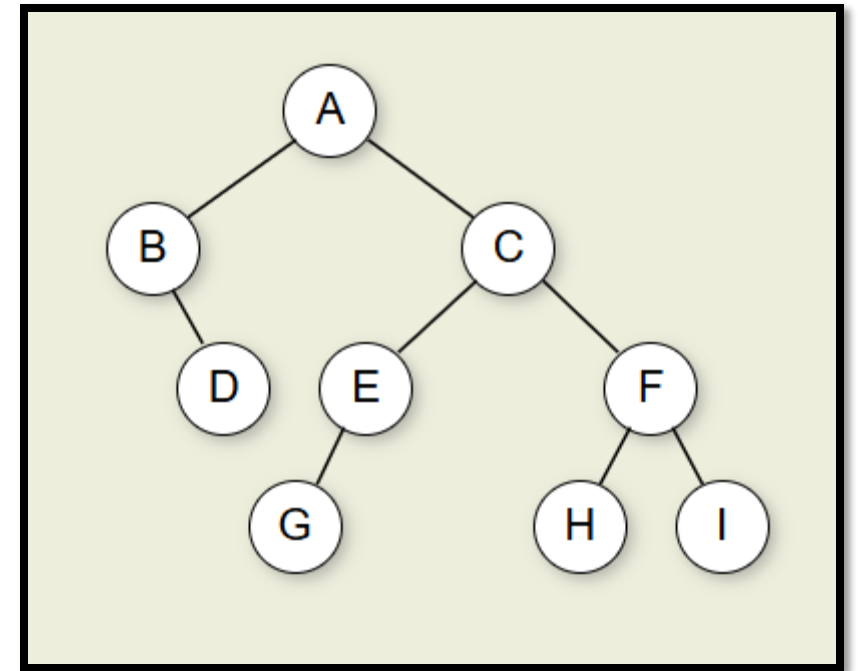


Postorder: 4, 2, 7, 8, 5, 6, 3, 1

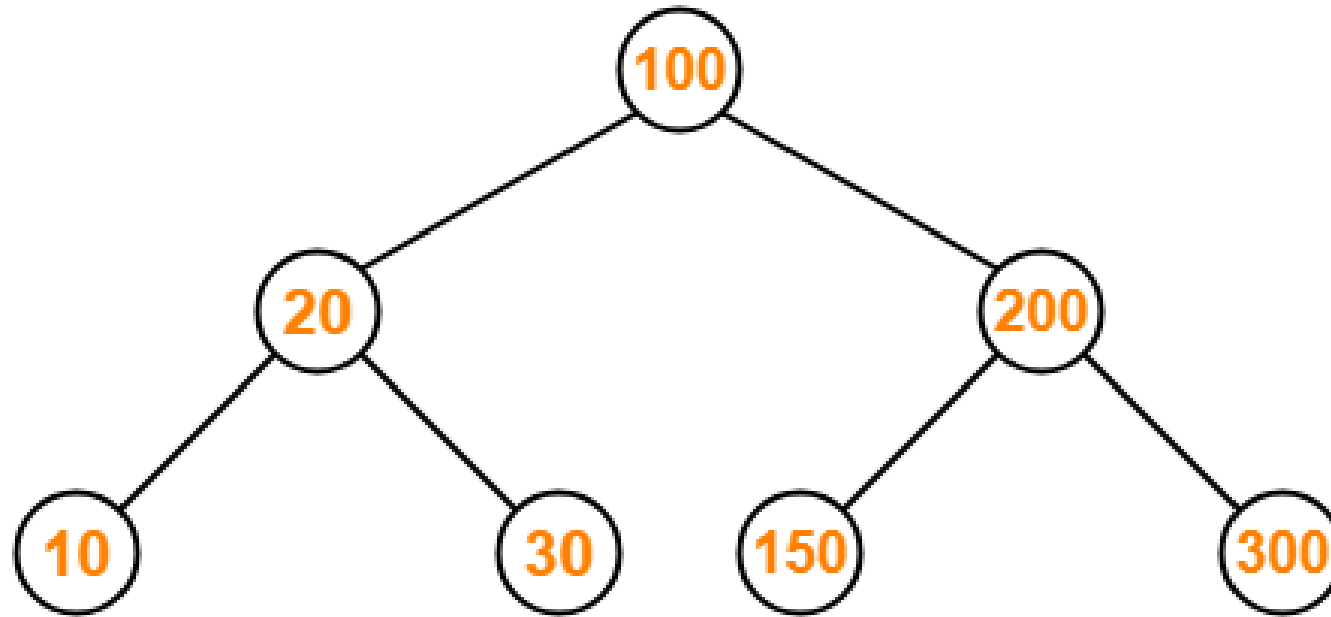
Tree Traversal: **Example**

What is the inorder traversal of the following binary tree

The preorder enumeration for the tree is
B G E H I F C A



Tree Traversal: Example



Binary Search Tree

Preorder Traversal-

100 , 20 , 10 , 30 , 200 , 150 , 300

Inorder Traversal-

10 , 20 , 30 , 100 , 150 , 200 , 300

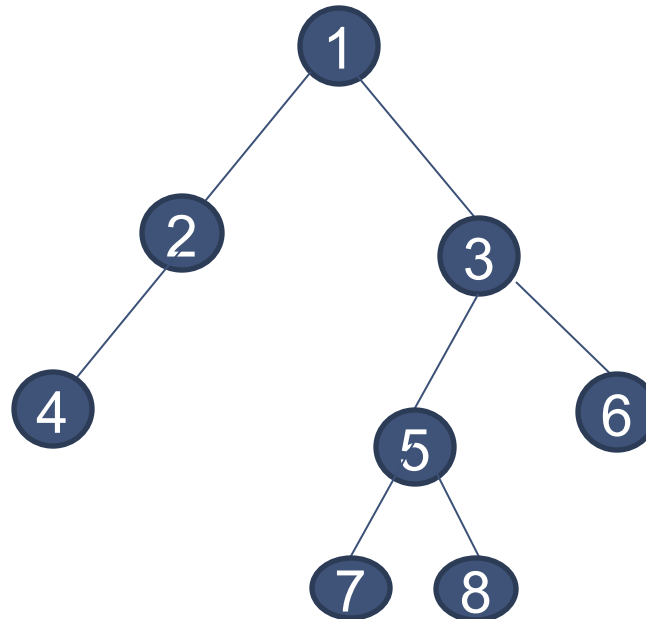
Postorder Traversal-

10 , 30 , 20 , 150 , 300 , 200 , 100

Tree Traversal: Example

Construct the binary tree if the preorder and inorder sequence is known.

Inorder Traversal: { 7, 5, 8, 3, 6 }
Preorder Traversal: { 3, 5, 7, 8, 6 }



Tree Traversal: Example

Consider the following Inorder and Postorder traversal of a binary tree:

Inorder = {D, B, E, A, F, G, C, H}

Postorder = {D, E, B, F, G, H, C, A}

What is the preorder traversal of the constructed tree

A, B, D, E, C, G, F, H

Tree Traversal: Example

The postorder traversal of a binary tree is 8, 9, 6, 7, 4, 5, 2, 3, 1.
The inorder traversal of the same tree is 8, 6, 9, 4, 7, 2, 5, 1, 3.
The height of a tree is the length of the longest path from the root to any leaf. The height of the binary tree above is

- A. 2
- B. 3
- C. 4
- D. 5

